

Day 12: Essential DAX Aggregation & Logical Functions

📌 Definition to Remember

Essential DAX functions provide standard mathematical aggregations (**SUM** , **AVERAGE** , **COUNT**), safe division (**DIVIDE**), and conditional branching logic (**IF** , **SWITCH**).

1. Core Mathematical & Aggregation Functions

```
// Sum and Average
Total Revenue = SUM(Fact_Sales[Revenue])
Average Order Value = AVERAGE(Fact_Sales[Revenue])

// Row Counts
Total Orders = COUNT(Fact_Sales[OrderID])
Unique Customers = DISTINCTCOUNT(Fact_Sales[CustomerID])
Total Rows = COUNTROWS(Fact_Sales)
```

2. Safe Mathematical Division: **DIVIDE()**

Always use **DIVIDE()** instead of the standard **/** operator to eliminate divide-by-zero crashes:

```
Profit Margin % =
DIVIDE(
    [Total Profit],
    [Total Revenue],
    0 // Alternate fallback result when denominator is 0 or null
)
```

3. Conditional Logical Branching: IF() vs SWITCH()

```
// Using IF for single condition
Discount Tier =
IF(
    [Total Sales] > 100000,
    "Platinum",
    "Standard"
)

// Using SWITCH(TRUE(), ...) for multi-condition evaluation (Cleaner than nested IFs)
Customer Performance Rating =
SWITCH(
    TRUE(),
    [Total Sales] >= 500000, "Tier 1: Enterprise Leader",
    [Total Sales] >= 100000, "Tier 2: Key Strategic Account",
    [Total Sales] >= 25000, "Tier 3: Growth Customer",
    "Tier 4: Standard Customer"
)
```

★ Must-Master Skills & Interview Points

1. Why `DIVIDE(A, B, 0)` is safer and faster than `IF(B = 0, 0, A / B)` .
2. Difference between `COUNT()` (ignores blanks), `COUNTROWS()` (table row count), and `DISTINCTCOUNT()` .
3. Mastering `SWITCH(TRUE(), ...)` for clean multi-tier business logic.

⚡ Quick Recall

`SUM / AVERAGE / DISTINCTCOUNT + DIVIDE(Safe Division) + SWITCH(TRUE(), conditions)`